

Multi-Agent System Design and Evaluation Report

1. Introduction

This assignment focused on designing and implementing a multi-agent workflow using CrewAI for a realistic business problem. The selected task was to analyze a customer-support dataset, validate its quality, identify major dissatisfaction drivers, and prepare a stakeholder-ready improvement brief. Instead of assigning the entire workflow to one general-purpose agent, three specialized agents were designed to handle different stages of the process.

The workflow consists of a **Data Quality Analyst**, a **Customer Insights Researcher**, and a **Stakeholder Communications Strategist**. This separation allows each agent to focus on a specific responsibility while passing structured outputs to the next agent.

2. Multi-Agent Design

The first agent, the **Data Quality Analyst**, is responsible for inspecting the customer-support data and identifying issues such as missing values, duplicates, inconsistent labels, and suspicious records. Its main goal is to ensure that downstream analysis is based on reliable information.

The second agent, the **Customer Insights Researcher**, uses the quality analyst's output as context and investigates dissatisfaction drivers and trends. It focuses on areas such as ticket category, sentiment, resolution time, and repeat contacts. The researcher is also instructed to distinguish measured evidence from hypotheses.

The third agent, the **Stakeholder Communications Strategist**, converts the validated findings into a concise business brief. Its responsibilities include preparing an executive summary, prioritizing recommendations, identifying owners, explaining expected impact, and suggesting a measurement plan.

Using multiple specialists is beneficial because the three stages require different types of reasoning. Data validation, analytical interpretation, and business communication have different objectives and quality requirements. A single generalist agent may be sufficient for a small and stable task, but specialized agents provide clearer responsibilities and more structured handoffs for a complex workflow.

3. Agents, LLMs, and Tools

Each agent was given its own LLM configuration and role-appropriate tools. The Data Quality Analyst uses a low-temperature configuration to favor consistent and deterministic analysis, while the Researcher uses a slightly higher temperature for analytical flexibility. The Communications Strategist uses a higher temperature to allow more flexibility in producing stakeholder-friendly language.

The **FileReadTool** is assigned to the Data Quality Analyst and Customer Insights Researcher because both require access to project data. The **FileWriterTool** is assigned to the Communications Strategist because its responsibility is to prepare the final business output. Restricting tools according to agent responsibilities prevents unnecessary access and keeps the workflow modular.

4. Sequential Workflow and Task Handoffs

The three tasks were implemented as CrewAI Task objects and connected using a sequential process. The quality task runs first, followed by the research task, and finally the communication task. The research task receives the quality task as context, while the communication task receives both previous outputs.

A potential handoff problem was identified during the design: the quality analyst could produce general prose, while the researcher required consistent information for further analysis. This was addressed by explicitly requiring the quality analyst to provide a structured Markdown table containing the issue type, affected records or fields, severity, evidence, and recommended handling. This makes the handoff more predictable and reduces ambiguity for the downstream agent.

Execution was attempted and the workflow generated execution logs. However, complete execution was limited by the unavailable dataset and Gemini API quota exhaustion. Therefore, the implementation and workflow structure could be evaluated, but a complete business analysis could not be produced from the actual dataset.

5. Hierarchical Delegation

A second CrewAI workflow was designed using `Process.hierarchical`. A separate **Crew Manager** was introduced to delegate work, review handoffs, identify missing evidence or malformed outputs, and ensure that the final result is coherent.

The sequential process provides a predictable workflow with explicit dependencies and relatively low coordination overhead. It is suitable for stable pipelines where each stage has a clearly defined output. The hierarchical process provides more flexibility because the manager can review and coordinate the specialists, making it more suitable for ambiguous or high-impact tasks. However, hierarchical delegation introduces additional coordination and API calls, which can increase latency, token usage, cost, and exposure to quota or rate-limit failures.

A reusable measurement function was also implemented to record latency, reliability, token usage, quality checks, errors, and the complete execution log for each process.

6. Evaluation and Cost Awareness

Three success criteria were defined for evaluating the final workflow:

1. **Factual grounding** – claims should be supported by the available project data and missing or unreliable information should be acknowledged.
2. **Completeness** – the output should contain findings, recommendations, caveats, owners, and next steps.
3. **Professional tone** – the final brief should be concise, actionable, and appropriate for stakeholders.

The notebook also includes functionality for extracting prompt, completion, and total token usage from CrewAI outputs and estimating approximate cost using configurable pricing. Because the Gemini quota prevented the complete comparison from being executed, actual token-based cost figures were not generated. The cost calculation therefore reports unavailable values rather than inventing measurements.

The manual evaluation indicates that hierarchical delegation can improve completeness through additional review, while a single-agent approach remains attractive for smaller and more predictable tasks.

7. Conclusion

This assignment demonstrated how a business problem can be divided among specialized AI agents and connected through structured task dependencies. The multi-agent design provides clearer separation of responsibilities, better-structured handoffs, and opportunities for independent review. However, these benefits come with additional complexity, API calls, token usage, and potential reliability issues.

Overall, the multi-agent approach is most valuable for **ambiguous, high-impact, or multi-stage tasks that benefit from specialization and review**. For small and stable tasks, a well-designed single agent may be more efficient and cost-effective. The implementation successfully demonstrates both sequential and hierarchical CrewAI architectures, although complete empirical comparison was limited by the available dataset and Gemini API quota.